BEYOND DETECTION:
An Informed Deployable Solar Flare Forecasting and Real Time X-Ray Flux Prediction System

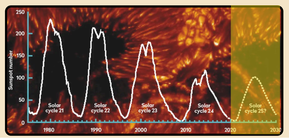
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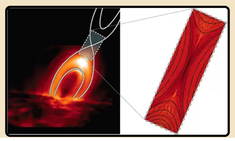
INTRODUCTION

What are Solar Flares?

- Explosions of electromagnetic energy that burst due to **magnetic reconnection**, the entanglement of plasma-filled loops on the Sun
- Classified into A-Class, B-Class, C-Class, M-Class, X-Class
  - Solar activity above C-1.0 in magnitude can cause **significant geomagnetic disturbances**
  - Radiation from X-Class and M-Class flares result in **radio blackouts** on Earth and **significant signal interference** with nearby satellites and space technology communication systems

Figure 1. Schwabe Cycle (NASA) **Figure 2.** Magnetic Reconnection (American Scholar)







Existing Flare Forecasting Techniques:

- Monitors and predicts flare activity within 24-48 hours
- **Statistical approaches:**
  - Failure due to poor generalization and limited assumptions
- **Deep Learning Techniques:**
  - CNNs and LSTMs
  - Require large datasets and lack physical constraints

PROBLEM STATEMENT

1. Focus on classification without forecasting energy emission evolution
2. Regression models often lack integrated magnetic and physical features
3. Data volume and computational cost limit scalability and operational efficiency

ENGINEERING OBJECTIVE

- Develop an **integrated flare forecasting framework** capable of:
  - Forecasting **continuous soft X-ray flux evolution,** the key indicator of flare strength, rather than just flare occurrence
  - Generating **comprehensive physics-informed reports** based on magnetic features and flux curves
  - Maintaining **computational and data efficiency**

INITIAL FOURIER ANALYSIS

Decomposed GOES X-ray flux into its frequency components

- Failed to produce distinct peaks at specific frequencies
- Indicates that flares are **impulsive broadband energy-release phenomena rather than periodic oscillations**
- Illustrates that prediction cannot be achieved through narrowband frequency detection and instead **requires analysis of nonlinear temporal evolution** or magnetic field properties

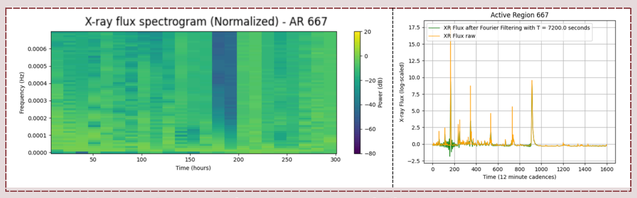
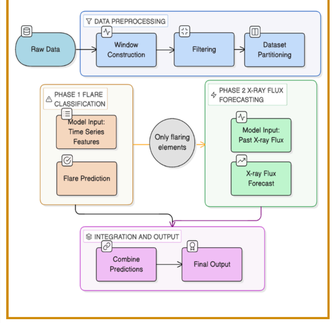


Figure 3-4. Fourier Analysis Spectogram and Filtering Comparison

METHODOLOGY

Figure 5. Methodology Flow Chart



Data Acquisition:

- Soft X-ray flux (0.1 to 0.8 nm) time series
- Spaceweather HMI Active Region Patch physical descriptors

Input/Output Construction:

- Constructed fixed-length input feature windows and varying output horizons
- Multi-class labels assigned

Cleaning and Filtering:

- Quality flag filtering of invalid observations
- Maintained within 70° from solar central meridian
- Log transformation and scaling
  - Computed using only the training set to prevent information leakage
- Removal of NaN and physically-invalid values

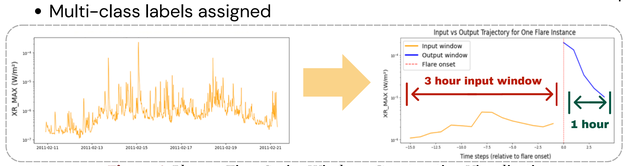


Figure 6. Phase 2 Time Series Windows Construction Visualization

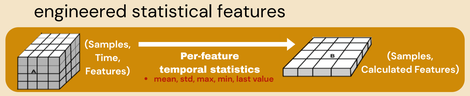
DATASET SUMMARY

Total samples: 7,976
Non flare class: 62%
M-flare class: 36%
X-flare class: 2.26%

Phase 1: Classification

Temporal Feature Aggregation:

- Flattened time series into a 2D dataset of engineered statistical features



XGBoost Model:

- Surpassed basic LSTM
- Model hyperparameters:
  - n_estimators = 100
  - learning rate = 0.1
  - max_depth = 6

Feature Importance:

$$Gain = \frac{1}{2}\left(\frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{(G_L+G_R)^2}{H_L+H_R+\lambda}\right) - \gamma$$

- Quantified total metric gain of each physical feature

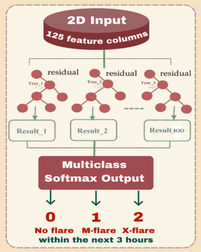


Figure 7. XGBoost model

Phase 2: Regression

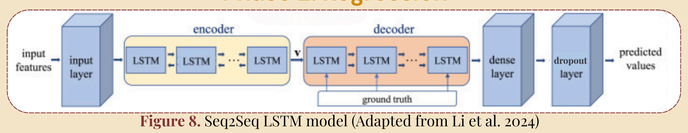


Figure 8. Seq2Seq LSTM model (Adapted from Li et al. 2024)

- Seq2Seq LSTM
- Attention-based Concatenation
- Huber Loss Optimization
- Early Stopping
  - Learning Rate Reduction
- Autoregressive Inference Model

Training-Specific Implementations

Teacher Forcing	Grid Search
• iteratively fed ground-truth target sequence	• Optimized hyperparameters • 5-fold cross-validation

Gaussian Noise Augmentation

- Addresses issue of sparse flare data
- Improves robustness to uncertainty

$$x' = x + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$
$$\sigma = 0.02$$

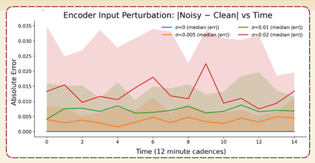


Figure 9. Gaussian noise on encoder inputs

RESULTS

Phase 1 Performance

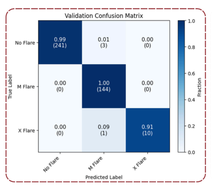


Figure 10. Confusion Matrix

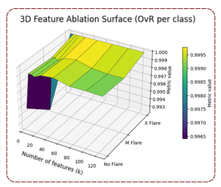


Figure 11. 3D Feature Ablation

Metrics

Overall accuracy:
- 99%

No flare-class:
- 99%

M-class
- 100%

X-class
- 91%

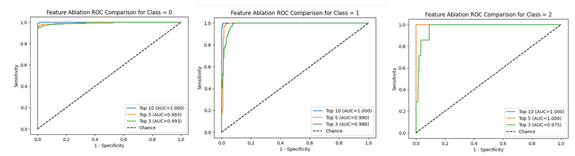


Figure 12-14. Feature Ablation ROC Comparisons for each class.

XGBoost Physical Feature Importance

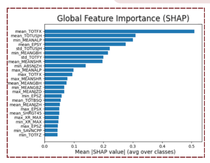


Figure 15. Global Feature Importance ranked on Shapley Additive Explanations values.

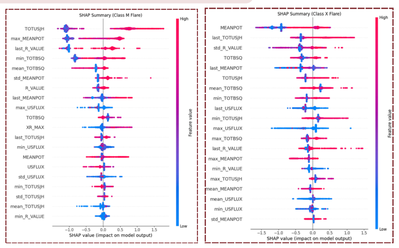


Figure 16-17. M- and X-class Feature Impact Summary.

Phase 2 Performance

M Flares

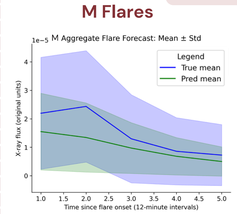


Figure 18. M Flare Testing Forecast

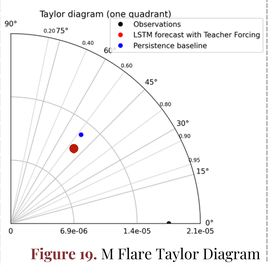


Figure 19. M Flare Taylor Diagram

X Flares

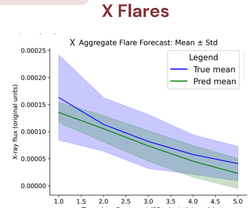


Figure 20. X Flare Testing Forecast

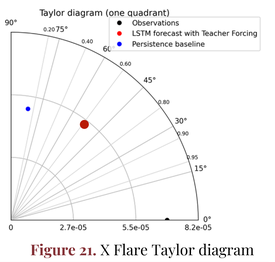


Figure 21. X Flare Taylor diagram

Metrics

X-class: RMSE = 5.56×10^{-5}, MAE = 3.73×10^{-5} (~16% of peak flux)
- $\log_{10}$ RMSE = 0.44

M-class: RMSE = 1.40×10^{-5}, MAE = 8.00×10^{-6} (~16% of peak flux)
- $\log_{10}$ RMSE = 0.38

Efficiency: Early stopping after 73 (X) / 68 (M) epochs with stable validation loss

Inference: 0.49 s (X) / 0.59 s (M) per unseen input

CONCLUSION AND DISCUSSION

Phase 1 – Classification:

- Achieved strong flare prediction with active-region physical parameters, **surpassing state-of-the-art approaches in accuracy** (~99%)
- ROC analysis across feature ablation levels indicated **high separability between flare classes** even with reduced feature sets

Phase 2 – Flux Forecasting:

- Seq2Seq LSTM with additive attention successfully **reconstructed flare X-ray flux trajectories with ~16% peak-magnitude error**
- Log-space RMSE values (0.38–0.44) indicate stable predictive performance across multiple orders of magnitude in flux intensity
- Taylor diagrams show that the algorithm's predictions **maintain higher correlation and lower variance error relative to the persistence baseline**, demonstrating improved agreement between predicted and observed flare evolution

Feature Importance & Physical Interpretation:

- Identified **total unsigned flux, Lorentz force proxies, and magnetic helicity parameters** as the strongest predictors of flare activity
- Correspond to magnetic energy accumulation and stress in solar active regions, supporting current physical models of flare initiation

Overall Efficiency:

- Two-stage architecture **reduced unnecessary regression computations** by first filtering candidate flares
- Achieved autoregressive forecast inference times below ~0.6 seconds

Further Research:

1. **Spatial Information Integration:** Extend the framework to include full vector magnetogram imagery alongside SHARP parameter vectors to enable hybrid CNN-LSTM architectures that capture spatial magnetic structure
2. **Physics-Guided Learning:** Explore physics-informed networks and constraints based on magnetohydrodynamic principles to improve interpretability and physical consistency of flare evolution forecasts

IMPACT AND APPLICATIONS

1. Processes around **120 flare forecasts per minute,** enabling rapid real-time monitoring of solar activity
2. Identifies **high-risk active regions early** and focuses detailed forecasting only where flares are likely
3. Provides alerts **3 hours in advance and continuously** forecasts the next **hour of X-ray flux**
4. Supports applications in **space-weather forecasting** that can help protect satellites, communication systems, and power infrastructure

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